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ÉTAT

Connexion

The function will be used in the collaborative space **Eurodrone_Design** and **Eurodrone_Design_Internal** with the role *Leader*.

Alternate Item Definition App - What is it ?

The Alternate Item Definition App is an app in CAD Authoring which is used to define an attribute named "Instance Alternate" or "Reference Alternate", created to notify that a reference or an instance is replaceable by one or more others. These attributes concern :

- Instance Alternate :
 - EP
 - SA
 - STD

- Flexible Fastener

- Reference Alternate :
 - EP
 - SA

The icon of this app is :

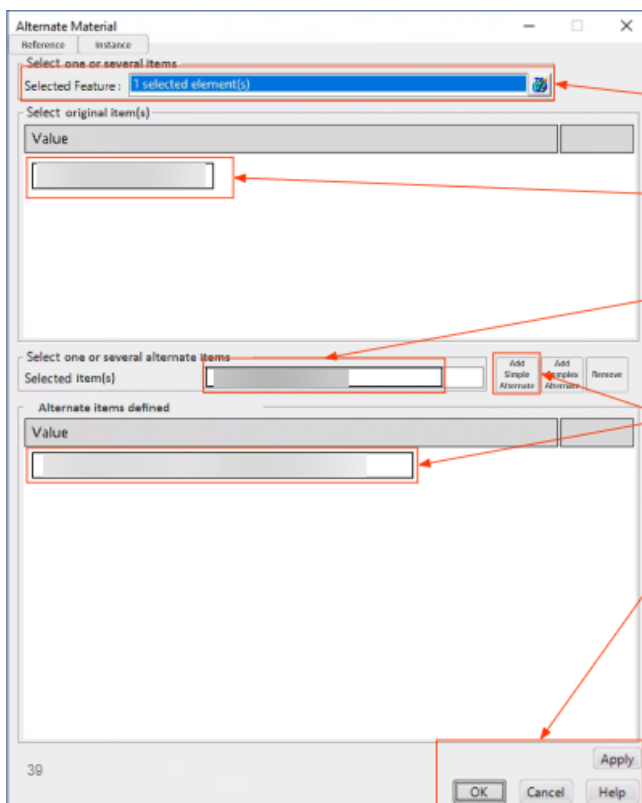


Presentation of the window's app

The window's app contains 2 different tabs, one for the instance and one for the reference. For each tabs, you have :

- 2 value framed
 - One to list the APNs of the objects before assigning Alternates
 - One to list the objects with the Alternates associated with them (already present or just added)
- A "Selected Feature" line, in order to select the object you want to work on in your product structure
- A "Selected Item(s)" line, in order to select the Alternate you want to associate to your objects
- A button "Simple Alternate", in order to link the chosen Alternate to the object. There are 2 use cases :
 - There is only one object in the first value frame : the Alternate selected in the "Selected Item(s)" line will be linked to the object selected in the "Selected Feature" line and this action will displayed a unique line in the second value framed corresponding to the information that will be set in the Attribute "Instance Alternate" or "Reference Alternate" in the Properties of the initial object
 - There are multiple object in the first value frame : the Alternate selected in the "Selected Item(s)" line will be linked to all the objects selected in the "Selected Feature" line and this action will displayed a line in the second value framed for each object corresponding to the information that will be set in the Attribute "Instance Alternate" or "Reference Alternate" in the Properties of the initial objects
- A button "Complex Alternate", in order to link multiple Alternate to one or multiple objects
- A button "Remove", in order to remove alternate already set in objects

The window look like the following picture :



Simple Alternate

Simple Alternate - Instance

In the Instance tab, you can add a simple alternate to these objects :

- EP
- SA
- Flexible Fastener
- STD

In the first value frame, the object is displayed this way : APN.Revision(InstanceTitle)

Value
Y471R0920-585.A(Y471R0920-585.1)

If there is only one object in the first value frame and one alternate selected with the "Selected Item(s)" line, the button "Complex Alternate" should be greyed.

Select original item(s)	
Value	
Y471R0920-585.A(Y471R0920-585.1)	
Y409D7825-305.A(Y409D7825-305.1)	

Select one or several alternate items	
Selected Feature :	
Y471R0920-585 A.1	

Alternate items defined

Add Simple Alternate Add Complex Alternate Remove

Once you've clicked on the button "Simple Alternate", the selected line in the first value frame should be duplicated in the second value frame and displayed that way : APN.Revision(AlternateAPN.Revision)

Alternate items defined	
Value	
Y471R0920-585.A(Y923E6142-929-FTA001.A)	
Y471R0920-585.A(Y923E6142-929-FTA001.A)	
Y471R0920-585.A(Y471R0920-585.A)	

You can also select multiple objects with the "Selected Feature" line. In that case, the button "Complex Alternate" should not be greyed and when you click on "Simple Alternate" after selecting multiple objects in the first value frame, all these objects are duplicated in the second value frame and displayed the same way. With this method, you will have one line for each selected object in the first value frame. However, you can only choose objects which come from the same product structure and with the same Business type with the "Selected Feature" line otherwise you'll have an error message.

When you set all your "Simple Alternate", you can click on "Apply" or "OK" to validate your modification and if you go in the Instance tab of your object's properties, you will find an attribute called "Instance Alternate" filled with the line which was displayed in the second value frame (ex. : Instance Alternate = Y71Z00346-200.A(Y12DD7454-260.A) with Y71Z00346-200=the initial EP and Y12DD7454-260=the Alternate). If you set multiple "Simple Alternate" on the same object, they will appear in the same attribute in the properties, separated by a " ; ".

Simple Alternate - Reference

In the Reference tab, you can add a simple alternate to these objects :

- EP
- SA

In the first value frame, the object is displayed this way : APN.Revision

<Image>

If there is only one object in the first value frame and one alternate selected with the "Selected Item(s)" line, the button "Complex Alternate" should be greyed.

<Image>

Once you've clicked on the button "Simple Alternate", the selected line in the first value frame should be duplicated in the second value frame and displayed that way : APN.Revision(AlternateAPN.Revision)

You can also select multiple objects with the "Selected Feature" line. In that case, the button "Complex Alternate" should not be greyed and when you click on "Simple Alternate" after selecting multiple objects in the first value frame, all these objects are duplicated in the second value frame and displayed the same way. With this method, you will have one line for each selected object in the first value frame. However, you can only choose objects which come from the same product structure **and** with the same Business type with the "Selected Feature" line otherwise you'll have an error message.

When you set all your "Simple Alternate", you can click on "Apply" or "OK" to validate your modification and if you go in the Instance tab of your object's properties, you will find an attribute called "Reference Alternate" filled with the line which was displayed in the second value frame (ex. : Reference Alternate = Y71Z00346-200.A(Y12DD7454-260.A) with Y71Z00346-200=the initial EP and Y12DD7454-260=the Alternate). If you set multiple "Simple Alternate" on the same object, they will appear in the same attribute in the properties, separated by a ",".

Complex Alternate

Complex Alternate - Instance

In the Instance tab, you can add a complex alternate to these objects :

- EP
- SA
- Flexible Fastener
- STD

In the first value frame, the object is displayed this way : APN.Revision(InstanceTitle)

If there is one or multiple objects in the first value frame and multiple alternate selected with the "Selected Item(s)" line, the button "Simple Alternate" should be greyed.

If only one object is displayed in the first value frame and at least 2 alternates are selected with the "Selected Item(s)" line, when you'll click on the "Complex Alternate" button, you'll have one line by alternate displayed in the second value frame with this notation : APN.Revision(AlternateAPN.Revision). However, if there are at least 2 objects selected in the first value frame **and** at least 2 Alternates selected with the "Selected Item(s)" line, there will be the exact same line displayed for each one of the objects in the first value frame with this notation : APNn°1.Revision,APNn°2.Revision,... (AlternateAPNn°1.Revision,AlternateAPNn°2.Revision,...). As for the "Simple Alternate", the objects selected in the first value frame must be part of the same product structure and have the same business type.

When you set all your "Complex Alternate", you can click on "Apply" or "OK" to validate your modification and if you go in the Instance tab of your object's properties, you will find an attribute called "Instance Alternate" filled with the line which was displayed in the second value frame

Example :

- Instance Alternate = Y71Z00346-200.A(Y12DD7454-260.A);Y71Z00346-200.A(Y95GX3504-180.A) with Y71Z00346-200 the initial EP and Y12DD7454-260 and Y95GX3504-180 the alternates.
- Instance Alternate = Y71Z00346-200.A,Y39J60001-300.A(Y12DD7454-260.A,Y95GX3504-180.A) with Y71Z00346-200.A and Y39J60001-300.A the initial EPs and Y12DD7454-260 and Y95GX3504-180 the alternates.

Complex Alternate - Reference

In the Reference tab, you can add a complex alternate to these objects :

- EP
- SA

In the first value frame, the object is displayed this way : APN.Revision

If there is one or multiple objects in the first value frame and multiple alternate selected with the "Selected Item(s)" line, the button "Simple Alternate" should be greyed.

If only one object is displayed in the first value frame and at least 2 alternates are selected with the "Selected Item(s)" line, when you'll click on the "Complex Alternate" button, you'll have one line by alternate displayed in the second value frame with this notation : APN.Revision(AlternateAPN.Revision). However, if there are at least 2 objects selected in the first value frame **and** at least 2 Alternates selected with the "Selected Item(s)" line, there will be the exact same line displayed for each one of the objects in the first value frame with this notation : APNn°1.Revision,APNn°2.Revision,... (AlternateAPNn°1.Revision,AlternateAPNn°2.Revision,...). As for the "Simple Alternate", the objects selected in the first value frame must be part of the same product structure and have the same business type.

When you set all your "Complex Alternate", you can click on "Apply" or "OK" to validate your modification and if you go in the Instance tab of your object's properties, you will find an attribute called "Reference Alternate" filled with the line which was displayed in the second value frame

Exemple :

- Reference Alternate = Y71Z00346-200.A(Y12DD7454-260.A);Y71Z00346-200.A(Y95GX3504-180.A) with Y71Z00346-200 the initial EP and Y12DD7454-260 and Y95GX3504-180 the alternates.
- Reference Alternate = Y71Z00346-200.A,Y39J60001-300.A(Y12DD7454-260.A,Y95GX3504-180.A) with Y71Z00346-200.A and Y39J60001-300.A the initial EPs and Y12DD7454-260 and Y95GX3504-180 the alternates.

Complex Alternate - Instance Fastener

Remove Button

On each tab Instance or Reference, you will find a button called "Remove". This button is there in order to remove alternates from objects.

To use it, you have to :

- Select the object(s) you want to modify with the "Selected Feature" line. The existing alternate will be automatically displayed in the second value frame.
- Select the alternate(s) you want to remove in the second value frame.
- Click on Remove, then Apply or OK

Once the procedure is validated and you go to the properties of the object you have just modified, the "Reference Alternate" or "Instance Alternate" attribute (depending on the tab in which you were) will no longer contain the Alternate that you just removed.

B.I. Essentials

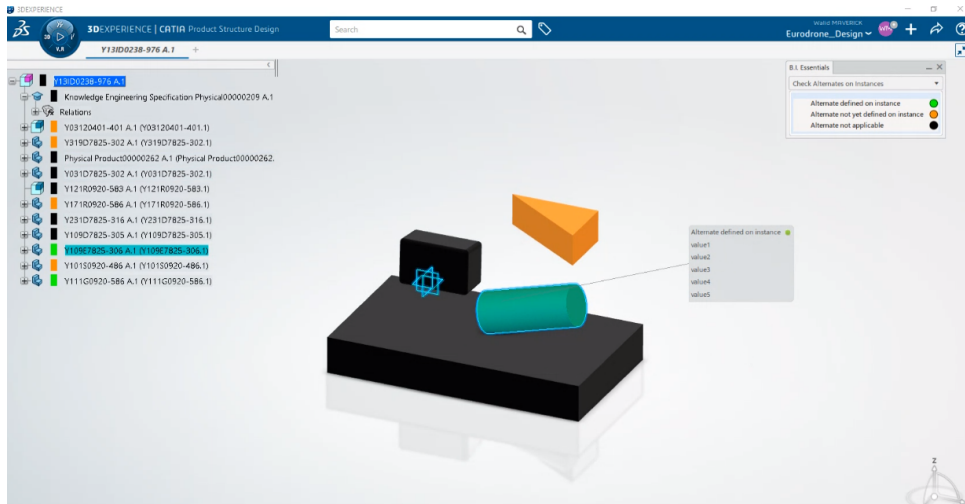
These B.I. Essentials had been created in order to see at the first look which object contains Instance Alternate and Reference Alternate.

We have 3 different colors which are displayed on the product structure and the geometry :

- Green : the object contain one or several alternates. The color is displayed on the product structure and on the geometry.
- Orange : the object may contain one or more alternates but does not contain any. The color is displayed on the product structure and on the geometry.

- Black : the object can't contain alternates. The color is displayed only on the product structure.

When one of these B.I. is opened, you can click on your object in the product structure or the geometry and that will open a pop-up containing the information of the respective attribute. For example, when you click on an object colored in green, the pop-up will contain the APN of this one as well as the different Alternates which have been assigned to it.



⚠ For **Instance Alternate** the colorization by BI essentials are displayed Only in the product structure and not in the geometry, for the Colorization in the 3D (Geometry) the Catalog **Eurodrone_Check_Alternate_on_Instances_Catalog** was implemented, story: [N2H-26122](#)

For lore information - [Instance Alternate Colorization Catalog](#)

Duplicate, New Revision and evolve Instance (N2H-24591)

Duplicate Action with Alternate Parts

At duplication, the attribute Alternate **will not be kept** on both instances and references

A warning message will be displayed to inform user that the values won't be kept in case the tool detects that one of the alternate attribute is not empty (on instance or reference)

Warning Message

“Alternates on Instances and References are not maintained after a duplicate. Please run B.I. Essentials on the original dataset to retrieve the original items including alternates.”

Video >>>>> [N2H-24591-Duplicate.mp4](#)

Revise Action with Alternate Parts

At revision, the attribute Alternate will be kept on both instances and references

Video >>>>> [N2H-24591-Revision .mp4](#)

Evolve Instance Action with Alternate Parts

At evolve instance (OOTB one) a BL **keep the value on Reference alternate**

Video >>>>> [N2H-24591- Evolve-1.mp4](#)

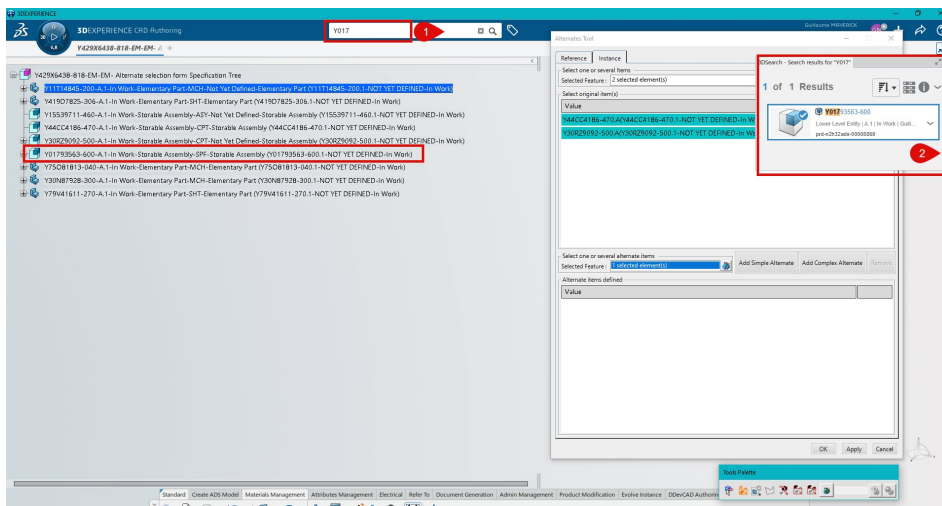
Alternate Selection from Specification Tree (N2H-26121)

The goal is to improve the current Alternate management tool, making it more user-friendly for end users, as well as taking advantage of its potential. It will aim to tackle the following issues:

- Selection of alternate items can be performed directly from the tree and not only from the 3DSearch bar.

Item selected by tree.mp4

On the other hand, if the user wishes, he will always have the choice of being able to carry out a search using the **3DSearch bar**.



Function's Availability & BL Involved:

The function **Alternate Item Management Command** is available in the CADAUTHORING app and developed by Maverick.

AIRBUS_Eurodrone_Alternate_Item_Definition_kcp is binded to **Eurodrone_Design** and **Eurodrone_Design_Internal** collab space

Is it Client/ Server/ Precommit/Know How Component?

Know How component

When it is triggered? Please specify the exact action

While launching Alternate Item Management command in CADAUTHORING APP

select Lower level entity on selected feature and select one or more several items from search, If there is only one object in the selected feature and select one or more several items then Add Simple alternate or if it is of multiple then Add Complex Alternate.

It will triggered when click on "OK".

the changes will reflected in the properties panel of attribute Instance Alternate and Reference Alternate.

What are the impacted objects?

Lower Level Entities impacted by

- Instance Alternate for :
 - EP
 - SA
 - STD
 - Flexible Fastener

- Reference Alternate for :
 - EP
 - SA

Instance Alternate and **Reference Alternate** are impacted in properties panel

Prefer to write technical names of attribute & ext

attributes:

AIRBUS_E_Identification_AirbusPartNumber

AIRBUS_E_BusinessType_BusinessType

AIRBUS_E_InstanceAlternate

AIRBUS_E_ReferenceAlternate

Technology used, etc.

EKL

The know how Alternate tool shall be updated with the story (N2H-28027)

- In the "Selected feature" the tool shall introspect all standards underneath the selected flexible fastener including the secondary fastener when existing
- The alternate information will be only put on the primary fastener instance
- We are not able to select the Secondary Fastener (Anchors & Nuts) as "**Selected Feature**"